

ELLIPTIC COMPLEMENTS OF CUBIC HYPERSURFACES

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ABSTRACT. Let $D \subset \mathbb{P}^n$, $n \geq 2$, be an arbitrary cubic hypersurface, and let D_{red} denote its reduced support. We prove that $\mathbb{P}^n \setminus D$ is holomorphically elliptic, and hence Oka, unless D_{red} is the union of three distinct hyperplanes containing a common codimension-two linear subspace. In the exceptional case, $\mathbb{P}^n \setminus D \cong (\mathbb{C} \setminus \{0, 1\}) \times \mathbb{C}^{n-1}$, so the complement is not Oka.

As applications, we prove that, for every elliptic curve E , the space of degree-three holomorphic maps $E \rightarrow \mathbb{P}^1$, and the space of degree-three holomorphic self-maps of $\mathbb{P}^1$, are both holomorphically elliptic, and hence Oka. For the second application, we identify the postcomposition quotient with the complement of an irreducible cubic hypersurface in $\mathbb{P}^4$.

1. INTRODUCTION

Oka theory grew from Oka's solution of the second Cousin problem on domains of holomorphy [50]. Grauert extended the principle to holomorphic bundles with homogeneous fibres over Stein spaces [29, 30, 31], and Gromov placed the theory within the h -principle and introduced elliptic manifolds via dominating holomorphic sprays [34, §§0.5–0.6]. Subsequent work developed Oka principles for maps and sections with interpolation along subvarieties, for submersions with sprays, and for subelliptic manifolds [13, 14, 15, 17, 18, 19]; see [20, 21, 28, 22, 25] for broader accounts. A central practical problem is to recognize the Oka property from concrete geometry, since its defining approximation properties rarely decide whether a given algebraic or quasi-projective variety is Oka. Dominating sprays, algebraic flexibility, density properties, and blowup constructions provide some of the main geometric mechanisms; see [16, 2, 41, 48, 42].

Hypersurface complements provide a particularly sharp testing ground: they lie at the interface with logarithmic hyperbolicity, where increasing degree is expected to produce the opposite behavior. In low degree, Hanysz studied Oka properties of hyperplane arrangements and meromorphic-graph complements [36, Theorems 3.1 and 4.6], as well as affine-conic configurations [37, Appendix B]. Kusakabe proved that every irreducible singular plane-cubic complement is Oka [45, Corollary 4.9(2), p. 1050]; see also [47, 27]. The remaining smooth plane cubic case was posed by Forstnerič–Lárusson [20, Problem B, p. 33], revisited in [21, §2.3.4, pp. 758–759], and remained recorded in [28, Remark 5.6, p. 394]. Buzzard and Lu obtained dominability by $\mathbb{C}^2$ [9, Proposition 5.1], but their map is ramified, and the Oka property does not in general descend through a ramified map.

These results leave a natural classification problem in degree three. Since the complement depends only on the reduced support of D , write D_{red} for the hypersurface obtained by discarding component multiplicities. Explicitly, if

$$F = c \prod_{j=1}^{N_F} F_j^{m_j}$$

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with $N_F \geq 1$ and distinct irreducible homogeneous factors F_j , then

$$D_{\text{red}} = \left\{ \prod_{j=1}^{N_F} F_j = 0 \right\}.$$

Theorem 1.1. *Let $D \subset \mathbb{P}^n$, $n \geq 2$, be a cubic hypersurface; no smoothness, irreducibility, or reducedness assumption is imposed. Then the following conditions are equivalent:*

- (i) $\mathbb{P}^n \setminus D$ is holomorphically elliptic;
- (ii) $\mathbb{P}^n \setminus D$ is Oka;
- (iii) D_{red} is not the union of three distinct hyperplanes containing a common linear subspace of codimension two.

Thus there is only one obstruction among cubic supports, and it is completely explicit. After a projective change of coordinates, the three hyperplanes have the form

$$Z_0 = 0, \quad Z_1 = 0, \quad Z_1 - Z_0 = 0.$$

Their complement is biholomorphic to

$$(\mathbb{C} \setminus \{0, 1\}) \times \mathbb{C}^{n-1},$$

which is not Oka. Indeed, on the chart $Z_0 \neq 0$, the coordinates $z = Z_1/Z_0$ and $w_j = Z_j/Z_0$, $2 \leq j \leq n$, identify the complement with the stated product: the first coordinate avoids 0 and 1, while the remaining coordinates are unrestricted.

The equivalence of the first two conditions follows from two general theorems of Gromov. Elliptic manifolds are Oka [34]; see also [17, Theorem 1.1, p. 529]. Conversely, these complements are affine, hence Stein, and every Stein Oka manifold is elliptic by Gromov's theorem [34]; see also [22, Proposition 5.6.15, p. 230]. We use the elliptic formulation because our proof constructs explicit dominating holomorphic sprays.

Two applications to mapping spaces.

The classification also yields two applications to natural mapping spaces in degree three.

Let E be an elliptic curve, and let $\mathcal{R}_d(E, \mathbb{P}^1)$ denote the complex manifold of degree- d holomorphic maps $E \rightarrow \mathbb{P}^1$. Bowman proved that, for $d = 3$, its Oka property is equivalent to that of the complement of a suitable smooth plane cubic [8, Corollary 30]. Together with Theorem 1.1, this gives the following stronger conclusion.

Theorem 1.2. *For every elliptic curve E , the manifold $\mathcal{R}_3(E, \mathbb{P}^1)$ is holomorphically elliptic. In particular, it is Oka.*

This settles the degree-three case of Forstnerič's Problem 7.6.23 [22, Problem 7.6.23, p. 347].

Let R_d denote the complex manifold of degree- d holomorphic self-maps of $\mathbb{P}^1$. Equivalently, a point of R_d is represented by a pair $[p : q]$ of binary forms of degree d without a common zero, modulo simultaneous nonzero scaling. Hanysz proved that R_3 is strongly dominable and strongly $\mathbb{C}$ -connected [37, Theorems 1.7 and 1.8], while its Oka property remained open.

Theorem 1.3. *The manifold R_3 is holomorphically elliptic. In particular, it is Oka.*

The connection with Theorem 1.1 comes from the postcomposition quotient of R_3 , which is biholomorphic to the complement of an irreducible cubic hypersurface in $\mathbb{P}^4$. The two applications are proved in Sections 8.2 and 8.3.

Proof strategy.

We describe the geometric mechanism behind Theorem 1.1. The smooth case is the essential one; singular and reducible cubics are handled separately.

Let $D = \{F = 0\} \subset \mathbb{P}^n$ be smooth. The cyclic cubic cover reduces the complement to a hyperplane complement in a smooth projective cubic. More precisely, $\{F = 1\}$ is an unramified cyclic cover of $\mathbb{P}^n \setminus D$, and

$$\{F = 1\} = S_F \setminus B, \quad S_F = \{F(x) = w^3\} \subset \mathbb{P}^{n+1}, \quad B = S_F \cap \{w = 0\}.$$

It remains to construct complement-preserving sprays on $S_F \setminus B$.

The proof has three separate tasks: to define the fibrewise flows for every complex time, to keep their trajectories in the hyperplane complement, and to obtain enough infinitesimal directions to dominate at every point. The common construction handles the first two tasks and the global compatibility of the local flows; the surface and higher-dimensional arguments establish domination by different geometric methods.

Residual conic fibrations and complement-preserving sprays.

A line on S_F determines a family of residual quadratic sections. Away from planes contained in S_F , these form a residual-conic fibration. Such fibrations are classical and occur in the compact setting in [43, Proposition 3.7 and Remark 3.8]; the point here is to globalize their fibrewise Hamiltonian flows while keeping every trajectory in the hyperplane complement. The local time parameters form a line-bundle cocycle, and identifying this cocycle allows the flows to glue. On cubic surfaces the 27 conic bundles give a finite dominating family. In higher dimensions the sprays first live on blowups along lines; a vanishing rescaling makes them descend, after which the tangent-transfer argument of Kaliman–Zaidenberg gives pointwise domination. We record the two cases separately.

Theorem 1.4. *Let $S \subset \mathbb{P}^3$ be a smooth cubic surface, let $H_0 \subset \mathbb{P}^3$ be a hyperplane, and assume that $B = S \cap H_0$ is smooth. Then the 27 lines on S furnish a finite global dominating family of rank-one holomorphic sprays on $S \setminus B$. In particular, $S \setminus B$ is holomorphically elliptic.*

Theorem 1.5. *Let $S \subset \mathbb{P}^{n+1}$ be a smooth cubic hypersurface of dimension $n \geq 3$, let $H_0 \subset \mathbb{P}^{n+1}$ be a hyperplane, and put $B = S \cap H_0$, which need not be smooth. Then $S \setminus B$ admits a finite global dominating family of holomorphic sprays on trivial line bundles. In particular, $S \setminus B$ is holomorphically elliptic.*

Integral singular cubics are reduced by projection from a singular point and Kusakabe’s localization theorem [45]; reducible and nonreduced cubics reduce to affine complements of degree at most two. These arguments are combined in Section 8.

Relation with compact cubic geometry.

The preceding construction is rooted in the classical geometry of lines on smooth projective cubic hypersurfaces; we refer to [39] for the general theory. Smooth cubic surfaces are uniformly rational [6, Example 2.4], and uniformly rational varieties are algebraically elliptic [3, Theorem 1.3]; see also [4] for related developments on rationality properties. Kaliman and Zaidenberg proved that every smooth cubic hypersurface of dimension at least two is algebraically elliptic [43, Theorem 1.1]. Their construction supplies the two geometric inputs used in our higher-dimensional argument: tangent directions of lines through a general point span the tangent space, and the third-point involution transports these directions to residual conics through an arbitrary point [43, Lemma 3.2, Proposition 3.7 and Remark 3.8]. These compact-cubic sprays need not preserve a hyperplane complement, since their images may cross the deleted hyperplane. We retain the tangent-spanning and tangent-transfer inputs, while the Hamiltonian construction above supplies the missing boundary control.

In a subsequent development, Guo and Xie used a related line-projection construction to prove that every smooth quartic hypersurface of dimension at least three is Oka [35].

The cubic theorem also lies at a natural numerical boundary for the logarithmic canonical class. For a smooth degree- d hypersurface $C \subset \mathbb{P}^n$, the logarithmic canonical class is

$$K_{\mathbb{P}^n} \otimes \mathcal{O}_{\mathbb{P}^n}(C) \cong \mathcal{O}_{\mathbb{P}^n}(d - n - 1).$$

For cubic hypersurfaces this bundle is trivial when $n = 2$ and anti-ample when $n \geq 3$. Such numerical behavior is related to logarithmic hyperbolicity questions, including the Green–Griffiths–Lang philosophy, which predicts algebraic degeneracy of entire curves in sufficiently positive complements [33]. Such numerical information alone does not imply the Oka property; the cubic result is proved by the explicit dominating sprays constructed above.

Forstnerič–Lárusson posed broader low-degree Oka questions [20, Question E, pp. 33–34]. Their asymptotic form leads to the following problem.

Question 1. Fix $d \geq 2$. For a general smooth hypersurface $H \subset \mathbb{P}^n$ of degree d , are H and $\mathbb{P}^n \setminus H$ Oka when n is sufficiently large relative to d ? What are the optimal relations between n and d in the two cases?

This is an opposite-direction analogue of the Kobayashi hyperbolicity problem, in both its compact and logarithmic forms: hyperbolicity is expected when the degree is large relative to the dimension, whereas the question above places the dimension high relative to the degree. The analogy is not a formal converse, and the thresholds for the compact hypersurface and its complement may differ. In degree three, a smooth plane cubic is a complex torus, while higher-dimensional smooth cubics are algebraically elliptic by Kaliman–Zaidenberg [43, Theorem 1.1]; hence the compact hypersurface is Oka for every $n \geq 2$. The present classification settles the complement in the same range. Thus the cubic case goes beyond the asymptotic range and includes the logarithmic Calabi–Yau boundary of a plane cubic complement.

The classification establishes holomorphic ellipticity, but the sprays produced by the proof involve matrix exponentials and are not shown to be algebraic. Recall that an algebraic variety is *algebraically elliptic* if it admits a dominating spray whose parameter bundle and spray map are algebraic. This leaves a finer question.

Question 2. Let $D \subset \mathbb{P}^n$, $n \geq 3$, be a smooth cubic hypersurface. Is $\mathbb{P}^n \setminus D$ algebraically elliptic?

The plane case has a negative answer, while Question 2 remains open; see Remark 8.2.

A related open problem asks whether every smooth algebraically elliptic n -fold admits a surjective algebraic morphism from $\mathbb{A}^n$; see [24, 5]. Kusakabe proved that one additional source dimension always suffices: every such n -fold admits a surjective algebraic morphism from $\mathbb{A}^{n+1}$ [46, Theorem 1.2].

Organization of the paper.

Section 2 recalls the Oka-theoretic tools, and Section 3 constructs the residual-conic sprays. Sections 4 and 5 prove Theorems 1.4 and 1.5, while Sections 6 and 7 handle reducible and integral singular cubics. Section 8 completes the classification: Section 8.1 proves Theorem 1.1; the two applications appear in Section 8.2 and Section 8.3.

2. ELLIPTICITY, SPRAYS, AND OKA PRINCIPLES

2.1. Sprays on complex manifolds. Let M be a complex manifold. A *holomorphic spray on M* is a triple (E, p, s) , where $p : E \rightarrow M$ is a holomorphic vector bundle and $s : E \rightarrow M$ is a holomorphic map satisfying

$$s(0_x) = x \quad (x \in M).$$

The vertical tangent space of E at 0_x is naturally identified with the fibre E_x . The restriction

$$(ds)_{0_x}|_{E_x} : E_x \longrightarrow T_x M$$

is called the *vertical differential* of the spray at x . The spray is *dominating* if this map is surjective for every $x \in M$.

A finite family of sprays (E_j, p_j, s_j) , $j = 1, \dots, m$, is *dominating* if

$$T_x M = \sum_{j=1}^m (ds_j)_{0_x}(E_{j,x}) \quad (x \in M).$$

The manifold M is *elliptic* if it admits a dominating holomorphic spray, and *subelliptic* if it admits a finite dominating family. See [22, Definition 5.6.13, pp. 229–230]; the elliptic notion originates in Gromov’s work [34, §0.5], and the subelliptic notion was introduced by Forstnerič [17].

The parameter bundles in these definitions need not be trivial. When a finite dominating family is defined on trivial bundles, its sprays can be composed successively to obtain one dominating spray [17, Lemma 2.1]. We use this composition in the higher-dimensional smooth case. On cubic surfaces the natural rank-one parameter bundles need not be trivial; there we first obtain subellipticity and then use the Stein converse below. Throughout the paper, every spray is defined globally on its full parameter bundle. An individual spray need not dominate at every point, but the finite families constructed below dominate jointly everywhere on their base manifold.

2.2. Consequences used in the proof. We record the Oka principles and localization results used below. For an algebraic variety, the corresponding algebraic notions are obtained by requiring the parameter bundles and spray maps to be algebraic. Kaliman–Zaidenberg proved that algebraic subellipticity and algebraic ellipticity are equivalent [44, Theorem 1.1].

Gromov proved that elliptic manifolds are Oka and, conversely, that Stein Oka manifolds are elliptic [34]. For projective manifolds, the converse was recently established by Forstnerič and Lárusson [26]. Forstnerič introduced subellipticity and proved

$$\text{elliptic} \implies \text{subelliptic} \implies \text{Oka};$$

see [17, Theorem 1.1, p. 529] and [22, Corollary 5.6.14 and Proposition 5.6.15, p. 230]. The complement of a projective hypersurface is affine and hence Stein. Consequently, the Oka property and holomorphic ellipticity are equivalent for the projective hypersurface complements considered here.

We shall also use the Oka principle for holomorphic fibre bundles [22, Theorem 5.6.5, p. 225] (see also [23, pp. 1–2]). Its special case for unramified holomorphic coverings is [22, Proposition 5.6.3, p. 224]. If a holomorphic fibre bundle has Oka fibre, then its total space is Oka if and only if its base is Oka. We apply this to line bundles, $\mathbb{C}^*$ -bundles, and finite coverings. A holomorphic retract of an Oka manifold is Oka as well; see [22, Proposition 5.6.8].

The singular case uses two additional inputs. The first is Kusakabe’s Zariski localization theorem.

Theorem 2.1. *Let M be a complex manifold. If every point of M has a Zariski-open Oka neighborhood in M , then M is Oka.*

This is [45, Theorem 1.4]. Here “Zariski-open” is understood in the holomorphic sense: the complement is a closed complex subvariety. We shall also use two consequences from the same paper: every projective quadric complement is Oka [45, Corollary 4.9(1)], and every integral singular plane-cubic complement is Oka [45, Corollary 4.9(2)].

The second input is Hanysz’s meromorphic-graph theorem [36, Theorem 4.6].

Following Hanysz, define the affine graph of a holomorphic map $m : X \rightarrow \mathbb{P}^1$ by

$$\Gamma_m = \{(x, m(x)) \in X \times \mathbb{C} : m(x) \neq \infty\}.$$

Thus, if $m(x) = \infty$, the graph has no point over x , and the entire affine fibre $\{x\} \times \mathbb{C}$ lies in its complement.

Theorem 2.2. *Let X be a complex manifold. If*

$$m = f + \frac{1}{g} : X \longrightarrow \mathbb{P}^1$$

for holomorphic functions f, g on X , then

$$(X \times \mathbb{C}) \setminus \Gamma_m$$

is Oka if and only if X is Oka.

3. THE COMMON RESIDUAL-CONIC CONSTRUCTION

Blowing up a line resolves the associated projection and produces an affine-plane family in which the residual cubic equation is quadratic. Its fibrewise Hamiltonian flow is parametrized by a line bundle. For cubic surfaces the blowup is trivial; in higher dimensions the resulting spray must later be descended to the original complement.

Let $S \subset \mathbb{P}^{n+1}$ be a smooth cubic hypersurface of dimension $n \geq 2$, let $H_0 \subset \mathbb{P}^{n+1}$ be a hyperplane, and put

$$B = S \cap H_0, \quad Y = S \setminus B.$$

For the surface theorem we assume that B is smooth; for $n \geq 3$, no smoothness is imposed on B . Fix a projective line $L \simeq \mathbb{P}^1 \subset S$ not contained in H_0 . Such a line exists as follows. If $n = 2$, the smooth cubic surface S has 27 lines [11, Chapter 9], and no one of them is contained in H_0 because $B = S \cap H_0$ is a smooth plane cubic. If $n \geq 3$, choose $x \in Y$ in the nonempty open line-spanning locus of [43, Lemma 3.2]. Since $x \notin H_0$, none of the lines through x is contained in H_0 ; choose any one of the lines supplied by that lemma. Write $L^\circ = L \cap Y$.

3.1. The ambient blowup and the affine conic. Let $L \subset \mathbb{P}^{n+1}$ be a projective line. The rational projection from L is not defined along L , and we resolve this indeterminacy by the blowup

$$b_L : Z_L = \text{Bl}_L \mathbb{P}^{n+1} \longrightarrow \mathbb{P}^{n+1}.$$

The exceptional divisor is

$$E_L = b_L^{-1}(L) \simeq \mathbb{P}(N_{L/\mathbb{P}^{n+1}}),$$

which records the normal directions to L and provides the missing data needed to extend the projection. Thus the rational projection from L extends to a morphism

$$p_L : Z_L \longrightarrow \mathbb{P}^{n-1}.$$

The fibre of p_L over a point of $\mathbb{P}^{n-1}$ is the plane spanned by L and the corresponding normal direction. In particular, p_L is a $\mathbb{P}^2$ -bundle.

Choose homogeneous coordinates $[z_0 : z_1 : y_2 : \cdots : y_{n+1}]$ in which

$$L = \{y_2 = \cdots = y_{n+1} = 0\}.$$

The planes through L are parametrized by $[u_2 : \cdots : u_{n+1}] \in \mathbb{P}^{n-1}$, and the blowup is the incidence variety

$$Z_L = \{([z_0 : z_1 : y], [u]) \in \mathbb{P}^{n+1} \times \mathbb{P}^{n-1} \mid y_i u_j = y_j u_i, \quad 2 \leq i, j \leq n+1\}.$$

The first projection is the blowdown map b_L , while the second projection is precisely the resolved projection p_L .

We use the convention that $\mathbb{P}(E)$ parametrizes one-dimensional subspaces of the fibres of E . Equivalently,

$$Z_L \cong \mathbb{P}_{\mathbb{P}^{n-1}}(\mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)),$$

where the fibre of the tautological bundle $\mathcal{O}(-1)$ at $[u]$ is the line $\mathbb{C}u$. Thus the fibre over $[u]$ is $\mathbb{P}(\mathbb{C}^2 \oplus \mathbb{C}u)$, namely the plane spanned by L and the normal direction $[u]$. Removing D_0 turns these projective planes into the affine planes carrying the residual quadratic equations.

Let H denote the divisor class of $b_L^* \mathcal{O}_{\mathbb{P}^{n+1}}(1)$, and let $\tilde{S}$ and D_0 denote the strict transforms of S and H_0 , respectively. Put

$$G_L = H - E_L.$$

The linear system of hyperplanes containing L defines the resolved projection p_L , and hence

$$\mathcal{O}_{Z_L}(G_L) = p_L^* \mathcal{O}_{\mathbb{P}^{n-1}}(1).$$

Since $L \not\subset H_0$, its hyperplane section has strict transform D_0 of class H . Moreover, S contains L with multiplicity one: otherwise a local equation of S would lie in $\mathcal{I}_L^2$, forcing all its first derivatives to vanish along L . Hence

$$(3.1) \quad [D_0] = H, \quad [\tilde{S}] = 3H - E_L = 2D_0 + G_L.$$

The ambient blowup restricts to $\tilde{S} \cong \text{Bl}_L S$. For $n = 2$, L is a Cartier divisor on S ; blowing up its invertible ideal does not change the surface, so $\text{Bl}_L S \cong S$. For $n \geq 3$, it is nontrivial. In all dimensions,

$$\tilde{Y}_L := \tilde{S} \setminus D_0 = \text{Bl}_{L^\circ} Y.$$

We write

$$\beta_L : \tilde{Y}_L \longrightarrow Y, \quad R_L = E_L \cap \tilde{Y}_L.$$

For $n = 2$, β_L is an isomorphism and R_L maps isomorphically onto L° ; for $n \geq 3$, R_L is the exceptional divisor.

Every fibre of p_L is a plane Π containing L . If $\Pi \not\subset S$, the restriction of the cubic equation of S to Π factors as the equation of L times a nonzero quadratic equation; the latter defines the residual conic, possibly singular or nonreduced. If $\Pi \subset S$, this quadratic equation vanishes identically. Since $L \not\subset H_0$, the intersection $\Pi \cap H_0$ is a line in Π . Hence D_0 restricts to a hyperplane on every $\mathbb{P}^2$ -fibre. Therefore,

$$\mathcal{A}_L := Z_L \setminus D_0$$

is an affine-plane bundle over $\mathbb{P}^{n-1}$, with fibres

$$\mathbb{P}^2 \setminus \mathbb{P}^1 \cong \mathbb{C}^2.$$

Here $Z_L \supset \mathcal{A}_L$ and $\tilde{S} \supset \tilde{Y}_L$ are respectively the ambient blowup and the transformed cubic with their boundary complements, while $\beta_L = b_L|_{\tilde{Y}_L}$ and $R_L = \beta_L^{-1}(L^\circ)$.

Let $s_{\tilde{S}}$ and s_{D_0} be defining sections of $\tilde{S}$ and D_0 . Since $\tilde{S} - 2D_0 \sim G_L$ and s_{D_0} is nowhere zero on $\mathcal{A}_L$, the quotient

$$q_L := \frac{s_{\tilde{S}}}{s_{D_0}^2} \in H^0(\mathcal{A}_L, p_L^* \mathcal{O}_{\mathbb{P}^{n-1}}(1))$$

is well defined. Its restriction to each affine-plane fibre is a quadratic polynomial, possibly the zero polynomial, and its zero set is precisely $\tilde{Y}_L$. If the corresponding plane Π is not contained in S , this zero set is the affine part of the residual conic. If $\Pi \subset S$, the entire affine plane is a fibre of $\tilde{Y}_L \rightarrow \mathbb{P}^{n-1}$.

In a local affine trivialization, q_L is a quadratic polynomial in the two fibre coordinates whose coefficients depend holomorphically on the base parameter.

3.2. Local coordinates and the Hamiltonian time line. The fibrewise quadratic equations provide the directions underlying the spray construction. We first describe these directions locally and then identify the line bundle that makes the corresponding Hamiltonian flows globally well defined.

Let (ξ_1, ξ_2) be affine coordinates on a fibre of

$$p_L : \mathcal{A}_L \longrightarrow \mathbb{P}^{n-1}.$$

For a quadratic equation

$$q(\xi_1, \xi_2) = 0,$$

the usual Hamiltonian vector field is

$$V_q = \frac{\partial q}{\partial \xi_2} \frac{\partial}{\partial \xi_1} - \frac{\partial q}{\partial \xi_1} \frac{\partial}{\partial \xi_2}.$$

Since $V_q(q) = 0$, this vector field is tangent to every level set of q , in particular to its zero set.

The formula depends on both the affine fibre coordinates and the local defining equation. Changing either multiplies the Hamiltonian field by a transition factor. Thus these local fields do not define an ordinary global vector field; instead, their time parameters must take values in a line bundle.

Set

$$\mathcal{Q}_L = p_L^* \mathcal{O}_{\mathbb{P}^{n-1}}(1)|_{\mathcal{A}_L}, \quad V_L = T_{\mathcal{A}_L/\mathbb{P}^{n-1}}.$$

Here $\mathcal{Q}_L$ is the line bundle in which the quadratic equation q_L takes values, while V_L is the relative tangent bundle of p_L : its fibre at $z \in \mathcal{A}_L$ is

$$(V_L)_z = \ker(dp_L)_z,$$

the tangent space to the affine-plane fibre through z . We denote their restrictions to $\tilde{Y}_L$ by $\mathcal{Q}_L|_{\tilde{Y}_L}$ and $V_L|_{\tilde{Y}_L}$.

Differentiating q_L only in the fibre directions gives the relative differential

$$d_{\text{rel}} q_L \in H^0(\tilde{Y}_L, V_L^\vee \otimes \mathcal{Q}_L).$$

Because V_L has rank two, its determinant line bundle identifies vectors and covectors fibre-wise. More precisely,

$$V_L^\vee \cong V_L \otimes (\det V_L)^\vee,$$

where the identification is induced by the alternating pairing

$$(v, w) \mapsto v \wedge w.$$

Hence

$$V_L^\vee \otimes \mathcal{Q}_L \cong \text{Hom}(\det V_L \otimes \mathcal{Q}_L^{-1}, V_L).$$

Under this identification, $d_{\text{rel}} q_L$ becomes the bundle morphism

$$(3.2) \quad \mathfrak{h}_L : \mathcal{N}_L \longrightarrow V_L|_{\tilde{Y}_L}, \quad \mathcal{N}_L := (\det V_L \otimes \mathcal{Q}_L^{-1})|_{\tilde{Y}_L}.$$

We call $\mathcal{N}_L$ the *Hamiltonian time line*: its fibres provide the parameters of the Hamiltonian flows. To see the construction explicitly, choose local frames ω of $\det V_L$ and e of $\mathcal{Q}_L$, and write $d_{\text{rel}} q_L = v \otimes e$. The vector $v = \mathfrak{h}_L(\omega \otimes e^{-1})$ is characterized by $v \wedge w = v(w)\omega$ for every $w \in V_L$. Taking $w = v$ gives $v(v) = 0$, so v is tangent to the zero locus of q_L . In fibre coordinates, this is the usual Hamiltonian vector $(q_{\xi_2}, -q_{\xi_1})$. Hence $d_{\text{rel}} q_L \circ \mathfrak{h}_L = 0$, and the image of $\mathfrak{h}_L$ is tangent to $\tilde{Y}_L$.

To identify its transition functions, we compute the class of the time line. Since L has codimension n in $\mathbb{P}^{n+1}$, the blowup formula gives

$$K_{Z_L} = b_L^* K_{\mathbb{P}^{n+1}} + (n-1)E_L = -(n+2)H + (n-1)E_L.$$

Moreover,

$$p_L^* K_{\mathbb{P}^{n-1}} = -nG_L = -n(H - E_L).$$

Subtracting these classes gives

$$(3.3) \quad K_{Z_L/\mathbb{P}^{n-1}} = -2H - E_L, \quad \det T_{Z_L/\mathbb{P}^{n-1}} = \mathcal{O}_{Z_L}(2H + E_L).$$

Since V_L is the restriction of the relative tangent bundle,

$$\det V_L = \mathcal{O}_{Z_L}(2H + E_L)|_{\mathcal{A}_L}.$$

Together with

$$\mathcal{Q}_L = \mathcal{O}_{Z_L}(H - E_L)|_{\mathcal{A}_L},$$

we obtain

$$(3.4) \quad \mathcal{N}_L = \mathcal{O}_{Z_L}(H + 2E_L)|_{\tilde{Y}_L}.$$

Since $E_L = H - G_L$, Eq. (3.4) may be rewritten as

$$\mathcal{N}_L = \mathcal{O}_{Z_L}(3H - 2G_L)|_{\tilde{Y}_L}.$$

On $\mathcal{A}_L = Z_L \setminus D_0$, the nowhere-zero section s_{D_0} trivializes $\mathcal{O}_{Z_L}(H)$. Dividing by $s_{D_0}^3$ therefore removes the $3H$ -term and gives

$$(3.5) \quad \mathcal{N}_L \xrightarrow{\sim} p_L^* \mathcal{O}_{\mathbb{P}^{n-1}}(-2)|_{\tilde{Y}_L}.$$

The three descriptions serve different purposes: the first defines the time parameter intrinsically, the second records its exceptional-divisor weight, and the third shows that its transition factors come from the base and hence remain constant along vertical trajectories. Thus the time line is pulled back from the base. In particular, its transition factors are constant along every Hamiltonian trajectory, which is the structural fact that allows the local flows to glue.

Choose $U_\alpha \subset \mathbb{P}^{n-1}$ over which $\mathcal{A}_L$ and $\mathcal{Q}_L$ are trivial. A point of $\mathcal{A}_L$ over U_α is written as (u, ξ_α) , where $u \in U_\alpha$ and $\xi_\alpha = (\xi_{\alpha,1}, \xi_{\alpha,2}) \in \mathbb{C}^2$ are affine fibre coordinates. Thus u denotes a base variable, ξ_α the two fibre variables, and τ_α below the Hamiltonian time coordinate.

These affine coordinates are not canonical: two local identifications of the same fibre with $\mathbb{C}^2$ may choose different origins. Hence the transition map on an overlap is generally affine rather than linear.

In a nowhere-zero frame e_α of $\mathcal{Q}_L$, write $q_L = q_\alpha e_\alpha$. Then q_α is quadratic in the fibre variables and

$$(3.6) \quad \tilde{Y}_L|_{U_\alpha} = \{q_\alpha(u, \xi_{\alpha,1}, \xi_{\alpha,2}) = 0\}.$$

A point $z \in \tilde{Y}_L$ is *relative regular* if $d_{\text{rel}} q_L(z) \neq 0$. At such a point, the scheme-theoretic zero set of q_L in the affine-plane fibre is a smooth curve near z . Points on a contained-plane fibre are not relative regular, even though that fibre is a smooth surface.

On overlaps $U_\alpha \cap U_\beta$, the coordinates and equations satisfy

$$(3.7) \quad \xi_\beta = A_{\beta\alpha}(u)\xi_\alpha + b_{\beta\alpha}(u),$$

$$(3.8) \quad q_\beta(u, \xi_\beta) = \lambda_{\beta\alpha}(u)q_\alpha(u, \xi_\alpha),$$

where $A_{\beta\alpha}(u)$ is an invertible holomorphic 2×2 matrix, while $b_{\beta\alpha}(u)$ records the change of origin, and $\lambda_{\beta\alpha}(u)$ is nowhere zero. In particular, the vertical differential of the coordinate change is $A_{\beta\alpha}(u)$; the translation term has zero vertical derivative.

3.3. The cocycle and gluing of the Hamiltonian flows. The transition factor has two sources: the change of the line-bundle-valued equation q_L and the change of the fibrewise area form. Their quotient is the cocycle of $\mathcal{N}_L$.

In the local coordinates of the previous subsection, the vertical Hamiltonian vector field associated with

$$q_\alpha(u, \xi_{\alpha,1}, \xi_{\alpha,2})$$

is

$$(3.9) \quad V_\alpha = (q_\alpha)_{\xi_{\alpha,2}} \frac{\partial}{\partial \xi_{\alpha,1}} - (q_\alpha)_{\xi_{\alpha,1}} \frac{\partial}{\partial \xi_{\alpha,2}}.$$

Let

$$\phi_{\beta\alpha}(u, \xi_\alpha) = (u, A_{\beta\alpha}(u)\xi_\alpha + b_{\beta\alpha}(u))$$

be the affine transition map between two local trivializations. The corresponding local equations satisfy

$$q_\beta(u, \xi_\beta) = \lambda_{\beta\alpha}(u)q_\alpha(u, \xi_\alpha).$$

We will show that

$$(3.10) \quad V_\beta = \frac{\lambda_{\beta\alpha}}{\det A_{\beta\alpha}} (\phi_{\beta\alpha})_* V_\alpha.$$

In Formula (3.10), $\lambda_{\beta\alpha}$ records the change of the local defining equation, while $\det A_{\beta\alpha}$ records the change of the fibrewise area form. Their quotient

$$\frac{\lambda_{\beta\alpha}}{\det A_{\beta\alpha}}$$

is the transition factor of $\mathcal{N}_L = \det V_L \otimes \mathcal{Q}_L^{-1}$; its reciprocal governs the Hamiltonian time parameter.

To verify the vector-field transformation formula, fix a base point u , and abbreviate

$$A = A_{\beta\alpha}(u), \quad b = b_{\beta\alpha}(u), \quad \lambda = \lambda_{\beta\alpha}(u).$$

We regard the fibre gradient

$$\nabla_{\xi_\alpha} q_\alpha$$

as a column vector. Differentiating

$$q_\beta(u, A\xi_\alpha + b) = \lambda q_\alpha(u, \xi_\alpha)$$

with respect to ξ_α , the chain rule gives

$$A^T \nabla_{\xi_\beta} q_\beta(u, A\xi_\alpha + b) = \lambda \nabla_{\xi_\alpha} q_\alpha(u, \xi_\alpha).$$

Hence

$$(3.11) \quad \nabla_{\xi_\beta} q_\beta(u, A\xi_\alpha + b) = \lambda A^{-T} \nabla_{\xi_\alpha} q_\alpha(u, \xi_\alpha),$$

where

$$A^{-T} = (A^{-1})^T$$

is the inverse transpose of A . It appears because the gradient is a covector, whereas tangent vectors transform by A .

Let

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The Hamiltonian vector field can be written in matrix form as

$$V_\alpha = J \nabla_{\xi_\alpha} q_\alpha.$$

For

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \Delta = \det A,$$

one has the elementary identity

$$JA^{-T} = \Delta^{-1} \begin{pmatrix} -a_{12} & a_{11} \\ -a_{22} & a_{21} \end{pmatrix} = \Delta^{-1} AJ.$$

Combining this identity with Eq. (3.11), and observing that the translation term b does not contribute to the vertical differential, we obtain

$$V_\beta(\phi_{\beta\alpha}(u, \xi_\alpha)) = \frac{\lambda}{\det A} AV_\alpha(u, \xi_\alpha),$$

which is exactly Eq. (3.10).

Set $\kappa_{\beta\alpha} = \lambda_{\beta\alpha} / \det A_{\beta\alpha}$, and denote the local flows by Φ_α^τ and Φ_β^τ . Since $\kappa_{\beta\alpha}$ depends only on the base parameter u , it is constant along every vertical Hamiltonian trajectory. Therefore

$$\Phi_\beta^\tau \circ \phi_{\beta\alpha} = \phi_{\beta\alpha} \circ \Phi_\alpha^{\kappa_{\beta\alpha}\tau}.$$

Equivalently, the local time coordinates satisfy

$$(3.12) \quad \tau_\beta = \frac{\det A_{\beta\alpha}}{\lambda_{\beta\alpha}} \tau_\alpha.$$

If local frames ν_α of the time line satisfy

$$\nu_\beta = \frac{\lambda_{\beta\alpha}}{\det A_{\beta\alpha}} \nu_\alpha,$$

then the same vector in a fibre is written $\tau_\alpha \nu_\alpha = \tau_\beta \nu_\beta$, which gives Eq. (3.12). The same identity ensures that the infinitesimal motions $\tau_\beta V_\beta$ and $\tau_\alpha (\phi_{\beta\alpha})_* V_\alpha$ agree on overlaps. This is the transition law of $\mathcal{N}_L = \det V_L \otimes \mathcal{Q}_L^{-1}$. The factors $\det A_{\beta\alpha} / \lambda_{\beta\alpha}$ satisfy the cocycle condition on triple overlaps and depend only on the base parameter. They are therefore constant along Hamiltonian trajectories, so the local flows glue on the total space of $\mathcal{N}_L$.

3.4. The entire flow. The Hamiltonian fields are affine-linear along the affine-plane fibres. Since the relative equation q_α is quadratic in the fibre coordinates, the local Hamiltonian field has the form

$$\dot{\xi} = M_\alpha(u) \xi + c_\alpha(u),$$

where $M_\alpha(u)$ is a 2×2 matrix and $c_\alpha(u)$ is a vector depending holomorphically on the base parameter u .

Affine-linear differential equations admit entire solutions in the complex time parameter. Explicitly, for $\tau \in \mathbb{C}$, the local flow is

$$(3.13) \quad \Phi_\alpha^\tau(u, \xi) = \left(u, e^{\tau M_\alpha(u)} \xi + \int_0^\tau e^{(\tau-s)M_\alpha(u)} c_\alpha(u) ds \right).$$

Since the integrand is entire in the integration variable, the integral is path independent and depends holomorphically on all variables. Hence Eq. (3.13) defines the flow for every $\tau \in \mathbb{C}$. By the Hamiltonian identity

$$V_\alpha(q_\alpha) = 0,$$

the flow preserves the level sets of q_α . Hence each trajectory remains in the same fibre of $p_L|_{\tilde{Y}_L}$.

The construction takes place on $\tilde{Y}_L = \tilde{S} \setminus D_0$; no extension of the spray across D_0 is needed.

By Eqs. (3.10) and (3.12), these entire local flows glue to a global holomorphic spray

$$(3.14) \quad \Phi_L : \mathcal{N}_L \longrightarrow \tilde{Y}_L, \quad \Phi_L(0_z) = z.$$

Formula (3.13) involves only the matrix exponential and an integral; it never divides by the discriminant of the quadratic equation. Hence the spray remains holomorphic when a residual conic degenerates. For example, the smooth parabola $q = y - x^2$ gives

$$V = \frac{\partial}{\partial x} + 2x \frac{\partial}{\partial y}, \quad \Phi^\tau(x, y) = (x + \tau, y + 2\tau x + \tau^2).$$

The nodal conic $q = xy$ gives the complete field $V = x\partial_x - y\partial_y$, which vanishes at the node. If q_L vanishes identically on an affine-plane fibre, then so does its Hamiltonian field, and the spray is the identity on that fibre. The same matrix formula therefore remains valid for singular or nonreduced conics and for the identically zero quadratic.

Proposition 3.1. *For every $n \geq 2$ and every line $L \subset S$ not contained in H_0 , the map*

$$\Phi_L : \mathcal{N}_L \longrightarrow \tilde{Y}_L$$

is a holomorphic spray defined on all of $\mathcal{N}_L$. It is vertical for p_L , preserves every fibre of $p_L|_{\tilde{Y}_L}$, and at every relative regular point z satisfies

$$(d\Phi_L)_{0_z}(\mathcal{N}_{L,z}) = \ker d(p_L|_{\tilde{Y}_L})_z.$$

Proof. The local flows glue by Eqs. (3.10) and (3.12), and Eq. (3.13) shows that the resulting spray is entire in the time parameter, including over singular fibres. Since it is vertical and preserves q_L , it preserves every fibre of $p_L|_{\tilde{Y}_L}$. Its vertical differential at the zero section is the Hamiltonian morphism $\mathfrak{h}_L$. At a relative regular point, $p_L|_{\tilde{Y}_L}$ is a submersion whose kernel is the one-dimensional tangent space to the smooth fibre curve; the nonzero vector $\mathfrak{h}_L$ spans this kernel. $\square$

4. THE SURFACE CASE: THE 27 LINES

We first specialize the common construction to the borderline case $n = 2$. Thus $S \subset \mathbb{P}^3$ is a smooth cubic surface and $B = S \cap H_0$ is a smooth plane cubic. The 27 lines on S will produce conic-bundle sprays whose directions jointly span TY . Here $B \sim -K_S$, so $K_S + B = 0$; this is the logarithmic Calabi–Yau boundary case.

Since a smooth plane cubic is irreducible, the hyperplane H_0 contains none of the 27 lines on S . Since $B \sim H$ and $H \cdot L = 1$, the scheme-theoretic intersection $B \cap L$ has length one and is therefore a single reduced point. In particular, the affine part

$$L^\circ := L \setminus B$$

is isomorphic to $\mathbb{C}$.

4.1. Conic bundles associated with lines. Let $L \subset S$ be one of the 27 lines. Choose linear forms ℓ_0, ℓ_1 with $L = \{\ell_0 = \ell_1 = 0\}$, and write

$$F_S = \ell_0 \Theta_0 + \ell_1 \Theta_1,$$

for quadratic forms Θ_0, Θ_1 . These forms have no common zero on L , since such a point would be singular on S . Hence projection from L extends to the conic-bundle morphism

$$(4.1) \quad \begin{aligned} \pi_L : S &\longrightarrow \mathbb{P}^1, \\ \pi_L(a) &= [\ell_0(a) : \ell_1(a)] \quad (a \notin L), \\ \pi_L(a) &= [-\Theta_1(a) : \Theta_0(a)] \quad (a \in L). \end{aligned}$$

On S , the equation $\ell_0 \Theta_0 + \ell_1 \Theta_1 = 0$ shows that the limiting ratio $[\ell_0 : \ell_1]$ along L is $[-\Theta_1 : \Theta_0]$. Since Θ_0 and Θ_1 have no common zero on L , this formula is defined everywhere and proves that the projection extends across L .

A plane containing L cuts the cubic surface into the fixed line L and a residual conic. Hence the fibres of π_L have divisor class

$$H - L.$$

This is the classical conic bundle structure associated with a line on a smooth cubic surface.

The boundary curve B is a bisection because

$$B \cdot (H - L) = H \cdot (H - L) = 3 - 1 = 2.$$

Indeed, B contains no fibre component, so the restriction of π_L to B has degree equal to this intersection number. Thus every residual conic meets B in a scheme of length two; after removing this intersection, it becomes the affine conic carrying the spray of Theorem 3.1.

4.2. Two skew conic bundles dominate. Choose six pairwise skew lines on S and contract them. The resulting blowdown $S \rightarrow \mathbb{P}^2$ identifies S with the blowup of $\mathbb{P}^2$ at six points in general position. We use the standard divisor-class description of the 27 lines and their incidence relations; see [11, Chapter 9].

Let ℓ be the pullback of the class of a line in $\mathbb{P}^2$, and let $e_1, \dots, e_6$ be the exceptional curves. Then

$$\ell^2 = 1, \quad \ell \cdot e_i = 0, \quad e_i \cdot e_j = -\delta_{ij},$$

and

$$H = -K_S = 3\ell - \sum_{i=1}^6 e_i.$$

The 27 lines on S have divisor classes

$$(4.2) \quad e_i, \quad L_{ij} = \ell - e_i - e_j, \quad Q_i = 2\ell - \sum_{k \neq i} e_k.$$

Every line meets ten other lines and is skew to sixteen, while two skew lines have exactly five common transversals.

Let $L, M \subset S$ be skew lines. The two conic bundles define the morphism

$$\Psi_{L,M} := (\pi_L, \pi_M) : S \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1.$$

The fibre classes are $H - L$ and $H - M$, and therefore

$$(4.3) \quad (H - L) \cdot (H - M) = H^2 - H \cdot L - H \cdot M + L \cdot M = 1.$$

The two fibre classes are effective, and their intersection number is one; hence fibres from the two pencils cannot be disjoint. Thus every fibre of π_L meets every fibre of π_M , so $\Psi_{L,M}$ is surjective. Since general fibres meet in one point, it is birational.

To locate the failure of local invertibility, let Γ be a curve contracted by the birational morphism $\Psi_{L,M}$. The intersection form is negative definite on the exceptional locus of a birational morphism of smooth surfaces, so $\Gamma^2 < 0$. Since $K_S = -H$, adjunction gives

$$\Gamma^2 = 2p_a(\Gamma) - 2 + H \cdot \Gamma,$$

and $H \cdot \Gamma \geq 1$ since H is ample. Hence $p_a(\Gamma) = 0$ and $H \cdot \Gamma = 1$, so Γ is a line in the cubic embedding.

A line $N \neq L, M$ on S is contracted by $\Psi_{L,M}$ precisely when it is contained in fibres of both conic bundle structures. Since the fibre classes of π_L and π_M are $H - L$ and $H - M$, respectively, this is equivalent to

$$(H - L) \cdot N = (H - M) \cdot N = 0.$$

For every line $N \subset S$, one has $H \cdot N = 1$. Hence

$$(H - L) \cdot N = 1 - L \cdot N, \quad (H - M) \cdot N = 1 - M \cdot N.$$

Therefore the above condition is equivalent to

$$L \cdot N = M \cdot N = 1,$$

which means precisely that N intersects both L and M .

The two distinguished lines are not contracted: since $L \cdot M = 0$,

$$(H - M) \cdot L = (H - L) \cdot M = 1.$$

Consequently, the exceptional locus of $\Psi_{L,M}$ is the union of the lines meeting both L and M :

$$(4.4) \quad \text{Exc}(\Psi_{L,M}) = \bigcup_{\substack{N \subset S \text{ a line} \\ N \cdot L = N \cdot M = 1}} N.$$

Lemma 4.1. *For every point $a \in S$, there exist skew lines*

$$L, M \subset S$$

such that

$$\Psi_{L,M} : S \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1$$

is a local biholomorphism at a .

Proof. By (4.4), it suffices to choose skew lines L, M such that no line through a meets both of them.

A point of a smooth cubic surface lies on at most three lines, since all such lines are contained in the tangent-plane section, which is a plane cubic. We consider the possible configurations.

If a lies on no line, any skew pair works. If a lies on a unique line A , choose $L = A$ and any line M skew to A . Then no line through a can meet both L and M .

Suppose that a lies on exactly two lines A_1, A_2 . Among the sixteen lines skew to A_1 , at most nine meet A_2 , since A_2 meets ten lines in total and one of them is A_1 . We may therefore choose a line M skew to both A_1 and A_2 . With $L = A_1$, neither line through a meets both L and M .

It remains to consider the Eckardt case, where three lines A_1, A_2, A_3 pass through a . The tangent-plane section gives

$$A_1 + A_2 + A_3 \sim H.$$

Every other line meets exactly one of the A_i , and each A_i meets eight such external lines, meaning lines not among A_1, A_2, A_3 . Choose an external line L meeting A_1 . The skew pair L, A_2 has five common transversals: A_1 and four of the eight external lines meeting A_2 . The other four external lines meeting A_2 are therefore disjoint from L ; choose one of them as M . Thus

$$L \cdot A_1 = M \cdot A_2 = 1,$$

while all other intersections between $\{L, M\}$ and $\{A_1, A_2, A_3\}$ vanish. Thus none of the three lines through a meets both L and M .

In all cases we have chosen skew lines L, M such that $a \notin \text{Exc}(\Psi_{L,M})$. Since $\Psi_{L,M}$ is a birational morphism between smooth surfaces and is an isomorphism away from its exceptional locus, it is a local biholomorphism at a . $\square$

4.3. The finite family of surface sprays. For every line $L \subset S$, identify $\tilde{Y}_L$ with Y via β_L , and denote the transported spray of Theorem 3.1 by

$$(4.5) \quad s_L : \mathcal{N}_L \longrightarrow Y.$$

Under this identification, $p_L = \pi_L$.

Proposition 4.2. *For every line $L \subset S$, the map s_L is a rank-one holomorphic spray. At every point $a \in Y$ where π_L is a submersion,*

$$(ds_L)_{0_a}(\mathcal{N}_{L,a}) = \ker(d\pi_L)_a.$$

Thus its infinitesimal direction is tangent to the residual affine conic.

Proof. This follows directly from Theorem 3.1, since $p_L = \pi_L$ under the identification $\tilde{Y}_L = Y$. $\square$

Proof of Theorem 1.4. Let $s_L : \mathcal{N}_L \rightarrow Y$ be the spray associated with each of the 27 lines. For $a \in Y$, Theorem 4.1 gives skew lines L, M such that (π_L, π_M) is a local biholomorphism at a . Therefore

$$\ker(d\pi_L)_a + \ker(d\pi_M)_a = T_a Y.$$

By Theorem 4.2, these two lines are the infinitesimal directions of s_L and s_M . The 27 sprays consequently form a finite dominating family. Thus Y is subelliptic and Oka; being affine and Stein, it is elliptic by Section 2. $\square$

5. THE HIGHER-DIMENSIONAL CASE

In higher dimensions the common spray lives on a nontrivial blowup. For a line $L \subset S$ not contained in H_0 , Section 3 gives a spray

$$\Phi_L : \mathcal{N}_L \longrightarrow \tilde{Y}_L.$$

It need not be constant on the exceptional fibres of $\beta_L : \tilde{Y}_L \rightarrow Y$, so it does not descend directly. Rescaling the time parameter by a section vanishing on the exceptional divisor makes the modified spray fix that divisor pointwise and descend to Y . Kaliman–Zaidenberg’s tangent-transfer argument gives pointwise domination, and Noetherianity reduces the resulting family to a finite one.

5.1. Exceptional scaling and proper-fibre descent. Let $R_L = E_L \cap \tilde{Y}_L$ be the exceptional divisor. By Eq. (3.4),

$$\mathcal{N}_L = \mathcal{O}_{Z_L}(H + 2E_L)|_{\tilde{Y}_L}.$$

Hence the defining sections of D_0 and E_L give

$$(5.1) \quad \eta_L := s_{D_0} s_{E_L}^2|_{\tilde{Y}_L} \in H^0(\tilde{Y}_L, \mathcal{N}_L).$$

The exponent two is dictated by the $2E_L$ -term in the class of $\mathcal{N}_L$; it is not an extra vanishing requirement for descent. In particular, $\eta_L|_{R_L} = 0$. The modified spray

$$(5.2) \quad \hat{s}_L : \tilde{Y}_L \times \mathbb{C} \longrightarrow \tilde{Y}_L, \quad \hat{s}_L(z, t) = \Phi_L(z, t\eta_L(z)).$$

Here t is the coordinate on the trivial parameter line; multiplication by $\eta_L(z)$ turns it into the Hamiltonian time $t\eta_L(z) \in \mathcal{N}_{L,z}$. Moreover, η_L vanishes on R_L ; hence $\hat{s}_L$ fixes R_L pointwise and has the fibrewise constancy required for descent.

Lemma 5.1. *Let $\beta : \hat{X} \rightarrow X$ be the blowup of a smooth centre, with exceptional divisor E , and let $\Phi : \mathcal{N} \rightarrow \hat{X}$ be a holomorphic spray. If $\eta \in H^0(\hat{X}, \mathcal{N})$ vanishes along E , then*

$$\hat{s}(z, t) = \Phi(z, t\eta(z))$$

fixes E pointwise, and $\beta \circ \hat{s}$ is constant on every fibre of $\beta \times \text{id}_{\mathbb{C}}$.

Proof. For $z \in E$, the equality $\eta(z) = 0_z$ gives $\hat{s}(z, t) = \Phi(z, 0_z) = z$. Thus E is fixed pointwise. Every fibre of β is either a singleton or is contained in E ; hence $\beta \circ \hat{s}$ is constant on the fibres of $\beta \times \text{id}_{\mathbb{C}}$. $\square$

We apply this observation to $\beta_L : \tilde{Y}_L \rightarrow Y$ through the following analytic descent lemma.

Lemma 5.2. *Let $f : \hat{X} \rightarrow X$ be a proper bimeromorphic holomorphic map between complex manifolds, with X normal, and let T be a complex manifold. If a holomorphic map $G : \hat{X} \times T \rightarrow A$, where $A \subset \mathbb{C}^N$ is a closed complex subvariety, is constant on the fibres of $f \times \text{id}_T$, then it factors uniquely through a holomorphic map $g : X \times T \rightarrow A$.*

Proof. The product $X \times T$ is normal, and $f \times \text{id}_T$ is a proper bimeromorphic modification. Hence

$$(f \times \text{id}_T)_* \mathcal{O}_{\hat{X} \times T} = \mathcal{O}_{X \times T};$$

by the analytic Riemann extension theorem [32, Chapter 8, §1]. The coordinate functions of G therefore descend holomorphically to $X \times T$. They define a map to $\mathbb{C}^N$, whose image lies in A because the defining equations of A vanish after pullback. Uniqueness follows from surjectivity. $\square$

The variety $Y = S \cap (\mathbb{P}^{n+1} \setminus H_0)$ is closed in the affine chart $\mathbb{P}^{n+1} \setminus H_0 \cong \mathbb{C}^{n+1}$, and is therefore affine. Choose an algebraic closed embedding $\varepsilon : Y \hookrightarrow \mathbb{C}^N$. By Theorem 5.1, the map $\varepsilon \circ \beta_L \circ \hat{s}_L$ is constant on the fibres of $\beta_L \times \text{id}_{\mathbb{C}}$. Hence Theorem 5.2 yields the holomorphic spray

$$(5.3) \quad s_L : Y \times \mathbb{C} \longrightarrow Y, \quad s_L(y, 0) = y.$$

5.2. Algebraicity at zero time. The descended spray is holomorphic. To make its tangent-spanning locus Zariski open and later extract finitely many sprays, we also need its zero-time field to be algebraic. This stronger infinitesimal statement holds because the Hamiltonian direction is algebraic, even though the complete flow need not be.

Recall from Eq. (3.2) the Hamiltonian bundle morphism $\mathfrak{h}_L : \mathcal{N}_L \rightarrow V_L|_{\tilde{Y}_L}$, and define

$$(5.4) \quad W_L(y) = \left. \frac{\partial}{\partial t} \right|_{t=0} s_L(y, t).$$

On $Y \setminus L^\circ$, the map $\beta_L : \tilde{Y}_L \rightarrow Y$ is an isomorphism. Hence, for $z = \beta_L^{-1}(y)$,

$$(5.5) \quad W_L(y) = d(\beta_L)_z(\mathfrak{h}_L(\eta_L(z))).$$

The right-hand side is algebraic, since it is obtained from the algebraic section η_L , the algebraic Hamiltonian morphism $\mathfrak{h}_L$, and the algebraic blowup map β_L .

At every relative regular point $z \notin R_L$ of the residual affine conic, $\mathfrak{h}_L(\eta_L(z))$ is nonzero and spans the tangent line of the conic. Therefore W_L is nonzero and spans the corresponding tangent line at every regular point of the residual affine conic in $Y \setminus L^\circ$.

It remains to extend this algebraic vector field across L° . Since

$$Y = S \setminus B$$

is smooth, its tangent sheaf T_Y is locally free and hence reflexive. Moreover,

$$\text{codim}_Y L^\circ = n - 1 \geq 2.$$

For the inclusion

$$j : Y \setminus L^\circ \hookrightarrow Y,$$

reflexive extension gives

$$T_Y \cong j_*(T_Y|_{Y \setminus L^\circ});$$

see [38, Proposition 1.6]. Hence the algebraic vector field on $Y \setminus L^\circ$ extends uniquely to an algebraic section of T_Y on all of Y . This is the algebraic Hartogs principle for a reflexive sheaf across a subset of codimension at least two.

The algebraic extension and W_L agree on the dense open set $Y \setminus L^\circ$, hence everywhere. Since the descended spray fixes L° , the field W_L vanishes there. Thus W_L is algebraic on all of Y , as required for finite extraction.

Proposition 5.3. *For every line $L \subset S$ not contained in H_0 , there exists a global holomorphic spray $s_L : Y \times \mathbb{C} \rightarrow Y$ whose zero-time field W_L is algebraic and vanishes along L° . At every $y \in Y \setminus L^\circ$ where the residual affine conic is regular, $W_L(y)$ is nonzero and spans its tangent line.*

5.3. Tangent transfer and pointwise spanning. Pointwise domination follows from the tangent-transfer mechanism of Kaliman–Zaidenberg, which transports tangent directions of lines through a general point to residual conics through an arbitrary point.

We use [43, Lemma 3.2] in the following form. There exists a nonempty Zariski open subset $U \subset S$ such that, for every $x \in U$, one can choose lines

$$L_1, \dots, L_n \subset S$$

through x satisfying

$$(5.6) \quad T_x S = \text{Span}(T_x L_1, \dots, T_x L_n).$$

The transfer is provided by the third-point involution used by Kaliman–Zaidenberg [43, Corollary 3.6(c), Proposition 3.7]. In introducing this birational map, they refer to [6, Example 2.4] and [7]. For fixed $\rho \in S$, this rational involution ι_ρ sends a general point $p \in S$ to the residual third point of the line $\langle \rho, p \rangle \cap S$. Suppose that the secant $\ell = \langle x, y \rangle$ meets S in the reduced divisor $x + \rho + y$. For completeness, the local property needed below follows directly from projection. Let π_ρ denote projection from ρ . Since the intersection

$S \cdot \ell = x + \rho + y$ is reduced, the line ℓ is tangent to S at neither x nor y . Hence $d\pi_\rho$ is an isomorphism at both points. After shrinking neighborhoods, the holomorphic inverse function theorem identifies neighborhoods of x and y with a common neighborhood of the direction of ℓ . The transition between the two inverse branches is the third-point involution ι_ρ . Thus ι_ρ is locally biholomorphic at x and y . If

$$\Pi_i = \langle L_i, y \rangle, \quad \Pi_i \cap S = L_i + \Gamma_i,$$

then ι_ρ maps the germ of L_i at x to the germ of Γ_i at y : for $p \in L_i$ near x , the third point of $S \cap \langle \rho, p \rangle$ is $\iota_\rho(p) \in \Gamma_i$. Therefore

$$(5.7) \quad T_y \Gamma_i = d\iota_\rho(T_x L_i)$$

holds.

Lemma 5.4. *For every $y \in Y$, there exist lines*

$$L_1, \dots, L_n \subset S,$$

none contained in H_0 and none passing through y , such that the residual conics through y are regular at y and

$$T_y Y = \text{Span}(W_{L_1}(y), \dots, W_{L_n}(y)).$$

Proof. Fix $y = [a_0 : \dots : a_{n+1}] \in Y$, and let $F = 0$ be an equation of S . Write $\mathbb{T}_y S$ for the projective tangent hyperplane to S at y . For $x \neq y$, the line $\langle x, y \rangle$ is tangent to S at y precisely when $x \in \mathbb{T}_y S$. It is tangent at x precisely when $\sum_{j=0}^{n+1} a_j F_{x_j}(x) = 0$. Thus the two bad loci are $S \cap \mathbb{T}_y S$ and

$$P_y := S \cap \left\{ \sum_{j=0}^{n+1} a_j \frac{\partial F}{\partial x_j} = 0 \right\}.$$

Both loci are proper. The smooth cubic S cannot be contained in its tangent hyperplane $\mathbb{T}_y S$, so the first is proper. If $P_y = S$, then the quadratic polar $\sum a_j F_{x_j}$ would vanish on S . Since the homogeneous ideal of the irreducible cubic hypersurface S is generated by its cubic equation F , no nonzero quadratic can vanish on S ; the polar would therefore vanish identically on $\mathbb{P}^{n+1}$. After a linear change of coordinates this would give $F_{x_0} = 0$, making S a cone and therefore singular.

The sets U and Y are nonempty Zariski open subsets of the irreducible variety S , while the loci just excluded are proper closed subsets. Therefore we may choose

$$x \in U \cap Y \setminus ((S \cap \mathbb{T}_y S) \cup P_y \cup \{y\}).$$

Then $\ell = \langle x, y \rangle$ is tangent to S at neither endpoint. Moreover, $\ell \not\subset S$: otherwise its tangent direction at y would lie in $\mathbb{T}_y S$, so $x \in \mathbb{T}_y S$, contrary to the choice of x . Bézout's theorem gives total intersection multiplicity three. Nontangency at x and y makes both endpoint multiplicities equal to one, so the intersection is reduced, the residual point ρ is distinct from them, and

$$S \cdot \ell = x + \rho + y.$$

Let $L_1, \dots, L_n$ be the lines through x supplied by (5.6). None is contained in H_0 , since $x \notin H_0$, and none contains y , since otherwise $\ell = \langle x, y \rangle = L_i \subset S$.

For each i , set $\Pi_i = \langle L_i, y \rangle$ and write

$$\Pi_i \cap S = L_i + \Gamma_i,$$

where Γ_i is the residual conic through y . Since $\ell \subset \Pi_i$ but $\ell \not\subset S$, the plane Π_i cannot be contained in S . By the local projection argument above, the third-point involution ι_ρ is biholomorphic near x and y . Hence ι_ρ maps the germ of L_i at x biholomorphically to the germ of Γ_i at y , and

$$T_y \Gamma_i = d\iota_\rho(T_x L_i).$$

To check regularity at y , write $F|_{\Pi_i} = \ell_i q_i$, with $L_i = \{\ell_i = 0\}$ and $\Gamma_i = \{q_i = 0\}$. Since $\ell_i(y) \neq 0$,

$$d(F|_{\Pi_i})_y = \ell_i(y) dq_i(y).$$

Thus $dq_i(y) = 0$ would give $d(F|_{\Pi_i})_y = 0$, so $T_y \Pi_i \subset T_y S$, equivalently $\Pi_i \subset T_y S$. This contradicts the transversality at y of $\ell \subset \Pi_i$. Hence Γ_i is regular at y .

By Theorem 5.3, the vector field $W_{L_i}(y)$ spans $T_y \Gamma_i$. Hence

$$\begin{aligned} \text{Span}(W_{L_1}(y), \dots, W_{L_n}(y)) &= d\iota_\rho(\text{Span}(T_x L_1, \dots, T_x L_n)) \\ &= T_y S = T_y Y. \end{aligned}$$

This proves the lemma. $\square$

5.4. Finite extraction. For an ordered n -tuple $\mathbf{L} = (L_1, \dots, L_n)$, let

$$\Omega_{\mathbf{L}} = \{y \in Y : W_{L_1}(y) \wedge \dots \wedge W_{L_n}(y) \neq 0\}.$$

The wedge in the definition is an algebraic section of $\det T_Y$, so every $\Omega_{\mathbf{L}}$ is Zariski open. These sets cover Y by Theorem 5.4. A Noetherian space is quasi-compact; hence this open cover has a finite subcover. Hence there are an integer $N_0 \geq 1$ and tuples $\mathbf{L}^{(1)}, \dots, \mathbf{L}^{(N_0)}$ such that

$$Y = \Omega_{\mathbf{L}^{(1)}} \cup \dots \cup \Omega_{\mathbf{L}^{(N_0)}}.$$

The sprays associated with all lines occurring in these tuples form a finite dominating family.

Proof of Theorem 1.5. By the composition lemma [17, Lemma 2.1], this family combines into a single dominating spray. Hence Y is holomorphically elliptic. $\square$

6. AFFINE QUADRATIC COMPLEMENTS

We next treat reducible cubics. A linear factor places their complements in an affine chart, where the remaining equation has degree at most two. Let q be a nonzero polynomial in $z_1, \dots, z_n$ of degree at most two, and put $A_q = \mathbb{C}^n \setminus \{q = 0\}$. Hanysz treated smooth affine plane conics [37, Appendix B], and Kusakabe proved that projective quadric complements are Oka [45, Corollary 4.9(1)]. The classification also requires degenerate and nonreduced quadratic equations.

Theorem 6.1. *Let $n \geq 1$. The manifold A_q is elliptic, hence Oka, unless an affine change of coordinates and multiplication of q by a nonzero constant transform it into*

$$z_1(z_1 - 1).$$

In the exceptional case

$$A_q \cong (\mathbb{C} \setminus \{0, 1\}) \times \mathbb{C}^{n-1}$$

and A_q is not Oka.

Proof. Write

$$q = q_2 + \ell + a,$$

where q_2 is homogeneous quadratic, ℓ is linear, and $a \in \mathbb{C}$. By $\text{rank } q_2$ we mean the rank of the symmetric bilinear form associated to q_2 .

Set $k_q := \text{rank } q_2$, and assume first that $k_q \geq 2$. After a linear change of coordinates,

$$q_2 = z_1^2 + \dots + z_{k_q}^2.$$

Let $e_1, \dots, e_n$ be the standard coordinate vectors. The vectors

$$\begin{aligned} v_1 &= e_1 + ie_2, & v_2 &= e_1 - ie_2, \\ v_j &= e_1 + ie_j & (3 \leq j \leq k_q), \\ v_j &= e_j & (k_q < j \leq n) \end{aligned}$$

form a basis of $\mathbb{C}^n$, and every one is q_2 -isotropic: $q_2(v_j) = 0$.

For these isotropic directions, we apply the standard overshear construction. In Varolin's coordinate-free terminology, if V is a $\mathbb{C}$ -complete holomorphic vector field and $f \in \ker V^2$, then fV is a complete overshear vector field; see [51, §3]. The explicit formula for its flow was given by Andrist and Kutzschebauch [1, Lemma 3.3]. We spell out the specialization to $V = D_v$ and $f = q$, since it shows directly that the resulting flow preserves A_q . The point of the present argument is that the q_2 -isotropic vectors chosen above form a basis of $\mathbb{C}^n$; the corresponding overshears can therefore be combined into a dominating spray on A_q .

Fix a q_2 -isotropic vector v , let D_v denote differentiation in the constant direction v , and set $b_v = D_v q$. Since q has degree at most two, Taylor's formula gives

$$(6.1) \quad q(z + \sigma v) = q(z) + \sigma D_v q(z) + \sigma^2 q_2(v) = q(z) + \sigma b_v(z).$$

Thus q varies affinely along every line parallel to v . Applying the same argument to the affine-linear function b_v , we obtain

$$b_v(z + \sigma v) = b_v(z) + \sigma D_v^2 q(z) = b_v(z),$$

because $D_v^2 q = 2q_2(v) = 0$. Hence b_v is constant along each such line.

Consider the holomorphic vector field $V_v = qD_v$. Its integral curve through z has the form $z + \sigma(t)v$, where Eq. (6.1) reduces the flow equation to

$$\sigma'(t) = q(z) + b_v(z)\sigma(t), \quad \sigma(0) = 0.$$

Writing

$$\chi(w) = \begin{cases} (e^w - 1)/w, & w \neq 0, \\ 1, & w = 0, \end{cases}$$

the solution is $\sigma(t) = t\chi(tb_v(z))q(z)$. The function χ is entire; its value at the origin incorporates the case $b_v(z) = 0$. Consequently, V_v is complete and its flow is

$$(6.2) \quad \Phi_v^t(z) = z + t\chi(tb_v(z))q(z)v, \quad (z, t) \in \mathbb{C}^n \times \mathbb{C}.$$

Substituting Eq. (6.2) into Eq. (6.1), we obtain

$$(6.3) \quad \begin{aligned} q(\Phi_v^t(z)) &= q(z) + t\chi(tb_v(z))q(z)b_v(z) \\ &= \left[1 + tb_v(z)\chi(tb_v(z))\right]q(z) \\ &= e^{tb_v(z)}q(z), \end{aligned}$$

Here the last equality follows from $1 + w\chi(w) = e^w$. Since the exponential factor never vanishes, Eq. (6.3) shows that the flow preserves A_q .

Compose the flows belonging to the basis above:

$$s(z; t_1, \dots, t_n) = \Phi_{v_n}^{t_n} \circ \dots \circ \Phi_{v_1}^{t_1}(z).$$

Each flow preserves A_q , and $\Phi_v^0 = \text{id}$; hence their composition is a holomorphic spray $A_q \times \mathbb{C}^n \rightarrow A_q$ fixing the zero section. Its differential in the parameter variables at the zero section has columns $q(z)v_1, \dots, q(z)v_n$. Since $q(z) \neq 0$, these columns form a basis of $T_z A_q$. Thus s is dominating and A_q is elliptic.

It remains to consider $k_q \leq 1$; repeated factors are included in this analysis. Rank zero is the affine-linear case. In rank one, completing the square will reduce the complement either to a product with $\mathbb{C}^*$ or to the exceptional one-variable polynomial. If $q_2 = 0$, then q is constant or affine linear. Its complement is respectively $\mathbb{C}^n$ or $\mathbb{C}^* \times \mathbb{C}^{n-1}$. These manifolds are elliptic: the additive and multiplicative complex groups have global dominating sprays, and ellipticity is preserved by products.

Suppose $k_q = 1$. Completing the square and making a linear change in the remaining variables gives

$$q = z_1^2 + b_2 z_2 + \dots + b_n z_n + a.$$

If some $b_j \neq 0$, we may assume after rescaling that $b_2 = 1$. The triangular polynomial automorphism

$$(z_1, z_2, z_3, \dots, z_n) \mapsto (z_1, q, z_3, \dots, z_n)$$

has inverse obtained by solving $q = z_1^2 + z_2 + \sum_{j=3}^n b_j z_j + a$ for z_2 ; hence it identifies A_q with $\mathbb{C}^{n-1} \times \mathbb{C}^*$. If every $b_j = 0$, then $q = z_1^2 + a$. For $a = 0$, the complement is again $\mathbb{C}^* \times \mathbb{C}^{n-1}$. For $a \neq 0$, the polynomial has two distinct roots, and an affine rescaling gives $q = z_1(z_1 - 1)$.

Finally, $\mathbb{C} \setminus \{0, 1\}$ is not Oka by the classical classification of Oka Riemann surfaces; see [36, Section 2]. It is a holomorphic retract of $(\mathbb{C} \setminus \{0, 1\}) \times \mathbb{C}^{n-1}$, so the latter cannot be Oka. $\square$

Remark 6.2. The proof shows that every nonexceptional A_q is holomorphically flexible in the sense of [2, Definition 6.4]: at each point, the values of $\mathbb{C}$ -complete holomorphic vector fields span its tangent space. For $k_q \geq 2$, the fields $qD_{v_1}, \dots, qD_{v_n}$ do so; the remaining nonexceptional cases are $\mathbb{C}^n$ and $\mathbb{C}^* \times \mathbb{C}^{n-1}$. This is holomorphic rather than algebraic flexibility: the overshear fields have algebraic coefficients, but their flows are generally nonalgebraic and the fields need not be locally nilpotent.

Corollary 6.3. *Let $n \geq 2$, and let $D \subset \mathbb{P}^n$ be a cubic hypersurface whose defining polynomial is reducible; repeated factors are allowed. Then $\mathbb{P}^n \setminus D$ is holomorphically elliptic unless D_{red} consists of three distinct hyperplanes containing a common linear subspace of codimension two. In the exceptional case it is not Oka.*

Proof. The cubic equation has a linear factor, say $F = LQ$, where L is linear and Q has degree two; repetitions are allowed. The complement lies in the affine chart $\{L \neq 0\} \cong \mathbb{C}^n$. Setting $L = 1$ turns Q into a nonzero polynomial q of degree at most two and gives

$$\mathbb{P}^n \setminus D \cong A_q.$$

Apply Theorem 6.1.

To translate the exceptional normal form projectively, choose coordinates with $L = Z_0$. Homogenizing $q = z_1(z_1 - 1)$ gives

$$F = Z_0 Z_1 (Z_1 - Z_0).$$

Thus D_{red} consists of three distinct hyperplanes. Their common intersection is given by $Z_0 = Z_1 = 0$, a linear space isomorphic to $\mathbb{P}^{n-2}$. Conversely, after choosing one of any three such hyperplanes as infinity, the other two become parallel affine hyperplanes, whose product is affinely equivalent to $z_1(z_1 - 1)$. This agrees with the exceptional case of Hanysz's hyperplane-arrangement theorem [36, Theorem 3.1]. $\square$

7. PROJECTION FROM A SINGULAR POINT

For integral singular cubics, projection from a singular point lowers the dimension and yields a recursive reduction. The complement is covered by two Zariski-open pieces: a $\mathbb{C}^*$ -bundle over a quadric complement and a graph complement on a cubic-root cover. Kusakabe's localization theorem then promotes these local Oka descriptions to the whole complement.

Lemma 7.1. *Let $n \geq 2$, and let*

$$F(x_0, x) = x_0 q(x) + c(x)$$

be a nonzero homogeneous cubic on $\mathbb{C}^{n+1}$, where $x = (x_1, \dots, x_n)$, q is homogeneous quadratic, and $0 \neq c$ is homogeneous cubic. If

$$\mathbb{P}^{n-1} \setminus \{c = 0\}$$

is Oka, then $\mathbb{P}^n \setminus \{F = 0\}$ is Oka.

Proof. Let $P = \{x_0 = 0\} \cong \mathbb{P}^{n-1}$ be the hyperplane of directions from $p = [1 : 0 : \cdots : 0]$, and put $\mathcal{L} = \mathcal{O}_P(1)$. Let

$$\pi : \mathbb{P}^n \setminus \{p\} \longrightarrow P, \quad [x_0 : x] \longmapsto [x],$$

be the projection from p . It identifies $\mathbb{P}^n \setminus \{p\}$ with the total space of the line bundle $\mathcal{L} \rightarrow P$. Explicitly, $b = [x] \in P$ represents the line $\ell = \mathbb{C}x$, the fibre $\mathcal{L}_b$ is ℓ^* , and $[x_0 : x]$ corresponds to the functional $\varphi \in \ell^*$ satisfying $\varphi(x) = x_0$. This description is independent of the chosen representative x .

In a local frame of $\mathcal{L}$, denote the fibre coordinate by t . The removed hypersurface has equation

$$(7.1) \quad tq + c = 0.$$

Here q and c are the local coefficient functions of sections of $\mathcal{O}_P(2)$ and $\mathcal{O}_P(3)$. Consequently, $-c/q$, where defined, transforms as the coefficient of a section of $\mathcal{L} = \mathcal{O}_P(1)$: under $x \mapsto \lambda x$, the quotient scales by $\lambda^{3-2} = \lambda$.

For $U = \mathbb{P}^n \setminus \{F = 0\}$, set

$$P_q = P \setminus \{q = 0\}, \quad P_c = P \setminus \{c = 0\}, \quad U^{(q)} = U \cap \pi^{-1}(P_q), \quad U^{(c)} = U \cap \pi^{-1}(P_c).$$

Since p lies on the removed cubic, π is algebraic on U ; hence $U^{(q)}$ and $U^{(c)}$ are Zariski open in U .

Assume first that $q \not\equiv 0$. Over P_q , the removed set in Eq. (7.1) is the image of the holomorphic section $-c/q$ of $\mathcal{L}$. The fibrewise translation $t \mapsto t + c/q$ sends this section to the zero section and identifies $U^{(q)}$ with $\mathcal{L}|_{P_q}$ minus its zero section. Thus $U^{(q)} \rightarrow P_q$ is a holomorphic $\mathbb{C}^*$ -bundle. The base P_q is the complement of a projective quadric and is Oka by [45, Corollary 4.9(1)]. Hence $U^{(q)}$ is Oka.

Over P_c , take the cubic-root covering

$$(7.2) \quad M_c = \{x \in \mathbb{C}^n : c(x) = 1\} \longrightarrow P_c, \quad x \longmapsto [x].$$

Since c is homogeneous of degree three, Euler's identity gives

$$\sum_j x_j \frac{\partial c}{\partial x_j} = 3c.$$

On $M_c = \{c = 1\}$, the right-hand side equals 3. Hence the differential dc cannot vanish at any point of M_c , and M_c is smooth. Scalar multiplication by μ_3 acts freely, with quotient P_c ; hence $M_c \rightarrow P_c$ is an unramified three-sheeted covering. Since P_c is Oka, so is M_c .

The pullback of $\mathcal{L}$ to M_c is trivial. The normalized representative x , for which $c(x) = 1$, trivializes this pullback by evaluation: it sends $\varphi \in (\mathbb{C}x)^*$ to the scalar $t = \varphi(x)$. In this trivialization the pullback of $U^{(c)}$ is

$$(7.3) \quad \mathcal{W} = \{(x, t) \in M_c \times \mathbb{C} : 1 + tq(x) \neq 0\}.$$

It is the complement of the affine part of the graph of the meromorphic function

$$m = -\frac{1}{q} : M_c \longrightarrow \mathbb{P}^1.$$

This function has Hanyasz's required form $f + 1/g$, with $f = 0$ and $g = -q$. At a zero of q , the value of m is infinity and the entire affine fibre belongs to $\mathcal{W}$. Hence Theorem 2.2 applies and shows that $\mathcal{W}$ is Oka.

The map $\mathcal{W} \rightarrow U^{(c)}$, $(x, t) \mapsto [t : x]$, has deck action $(x, t) \mapsto (\zeta x, \zeta t)$, $\zeta \in \mu_3$. This action is free: since $x \neq 0$, the equality $\zeta x = x$ forces $\zeta = 1$. Moreover, $1 + tq(x)$ is invariant because $q(\zeta x) = \zeta^2 q(x)$. Hence $\mathcal{W} \rightarrow U^{(c)} = \mathcal{W}/\mu_3$ is an unramified finite covering. Since the Oka property descends under such coverings, $U^{(c)}$ is Oka.

The two Zariski-open subsets $U^{(q)}, U^{(c)} \subset U$ cover U . Indeed, if $q(x) = c(x) = 0$, then $tq(x) + c(x) = 0$ for every point of the projection fibre. The whole fibre belongs

to the removed cubic and contributes no point to U . Kusakabe's localization theorem, Theorem 2.1, now gives that U is Oka.

If $q \equiv 0$, then P_q and $U^{(q)}$ are empty, while $U = U^{(c)}$. Equation Eq. (7.1) is then independent of the fibre coordinate, and U is the total space of $\mathcal{L}|_{P_c} \rightarrow P_c$. It is Oka by the Oka principle for holomorphic fibre bundles. $\square$

Theorem 7.2. *Let $D \subset \mathbb{P}^n$, $n \geq 2$, be an integral singular cubic hypersurface. Then $\mathbb{P}^n \setminus D$ is Oka and hence holomorphically elliptic.*

Proof. We argue by induction on n . The smooth input used below is Theorem 8.1; its proof depends only on Theorems 1.4 and 1.5 and is independent of the present induction. For $n = 2$, this is Kusakabe's theorem for integral singular plane cubics [45, Corollary 4.9(2)].

Assume $n \geq 3$ and choose $p \in \text{Sing } D$. Hyperplanes avoiding p form a nonempty open set in the dual projective space. By Bertini irreducibility [40, Théorème 6.3(4)], a general such hyperplane H has irreducible section $C = D \cap H$. We may also choose H transverse to the dense smooth locus of D , so C is generically reduced. As an effective Cartier divisor on the smooth variety H with irreducible support, C therefore has multiplicity one and is integral. Thus $C \subset H \cong \mathbb{P}^{n-1}$ is an integral cubic. Choose coordinates with

$$p = [1 : 0 : \cdots : 0], \quad H = \{x_0 = 0\}.$$

Write the defining cubic polynomial of D as

$$F = ax_0^3 + x_0^2\ell(x) + x_0q(x) + c(x).$$

Evaluating F at p gives $a = 0$. For $1 \leq j \leq n$, the value $F_{x_j}(p)$ is precisely the coefficient of x_j in ℓ . Since p is singular, all these coefficients vanish. Hence $\ell = 0$, so $F = x_0q + c$, and $c = F|_H$ defines C . If C is smooth, then $H \setminus C$ is Oka by Theorem 8.1. If C is singular, then $H \setminus C$ is Oka by the induction hypothesis. Therefore

$$\mathbb{P}^{n-1} \setminus C$$

is Oka in either case.

The singular-point reduction Theorem 7.1 now gives that $\mathbb{P}^n \setminus D$ is Oka. Since this complement is affine and Stein, Gromov's converse theorem makes it holomorphically elliptic. $\square$

8. CLASSIFICATION AND APPLICATIONS TO MAPPING SPACES

8.1. Proof of the classification theorem. We combine the smooth, reducible, and integral singular cases proved in the preceding sections.

Theorem 8.1. *The complement of every smooth cubic hypersurface in $\mathbb{P}^n$, $n \geq 2$, is holomorphically elliptic.*

Proof. Write $C = \{F = 0\}$ and consider the cyclic cubic cover

$$S_F = \{F(x) = w^3\} \subset \mathbb{P}^{n+1}, \quad B_F = S_F \cap \{w = 0\}.$$

Let $G(x, w) = F(x) - w^3$. Away from $w = 0$, the derivative $G_w = -3w^2$ does not vanish. On $w = 0$, a singular point of S_F would satisfy $F = 0$ and $dF = 0$, hence would give a singular point of C . Thus S_F is smooth. Moreover, $B_F \cong C$, while setting $w = 1$ identifies $S_F \setminus B_F$ with $\{F = 1\} \subset \mathbb{C}^{n+1}$. This affine hypersurface is elliptic by Theorem 1.4 for $n = 2$ and by Theorem 1.5 for $n \geq 3$.

The radial projection

$$\{F = 1\} \longrightarrow \mathbb{P}^n \setminus C, \quad x \longmapsto [x],$$

is an unramified three-sheeted covering. For each $[x]$ in the complement, the equation $\lambda^3 F(x) = 1$ has three solutions, and they form a free μ_3 -orbit. Hence $\mathbb{P}^n \setminus C$ is Oka and, being Stein, is elliptic by Section 2. $\square$

Proof of Theorem 1.1. Every cubic is either smooth, irreducible and singular, or reducible; a nonreduced cubic belongs to the last case. Thus Theorems 6.3, 7.2 and 8.1 show that every nonexceptional complement is Oka. Since it is affine and Stein, it is elliptic by [22, Proposition 5.6.15, p. 230].

In the exceptional case, $\mathbb{P}^n \setminus D \cong (\mathbb{C} \setminus \{0, 1\}) \times \mathbb{C}^{n-1}$. Its non-Oka first factor is a holomorphic retract, so the complement is neither Oka nor elliptic. $\square$

Remark 8.2. The method of this paper proves holomorphic ellipticity but does not establish algebraic ellipticity. If $D \subset \mathbb{P}^2$ is a smooth cubic, then $\mathbb{P}^2 \setminus D$ is not algebraically elliptic. Indeed, if $s : E \rightarrow \mathbb{P}^2 \setminus D$ is a dominating algebraic spray, then the maps $t \mapsto s(tv)$, with v ranging over the open subset on which the fibre differential at the zero section is nonzero, form a dominant family of nonconstant algebraic maps $\mathbb{A}^1 \rightarrow \mathbb{P}^2 \setminus D$. Thus the complement is separably $\mathbb{A}^1$ -uniruled in characteristic zero. By [10, Proposition 2.5 and Corollary 2.8], this contradicts $K_{\mathbb{P}^2} + D \cong \mathcal{O}_{\mathbb{P}^2}$. For a smooth cubic hypersurface $D \subset \mathbb{P}^n$ with $n \geq 3$, it is not known whether $\mathbb{P}^n \setminus D$ is algebraically elliptic.

8.2. Degree-three elliptic functions. We use the following elementary consequence of the Oka–Grauert principle.

Lemma 8.3. *Let $\pi : X \rightarrow B$ be a holomorphic principal G -bundle, where G is a complex Lie group. If B is Stein and holomorphically elliptic, then X is holomorphically elliptic.*

Proof. Let $s : V \rightarrow B$ be a dominating spray on a holomorphic vector bundle $p : V \rightarrow B$. The homotopy

$$H_t(v) = s(tv), \quad 0 \leq t \leq 1,$$

joins p to s , so the principal bundles p^*X and s^*X over V are topologically isomorphic. Since the total space V of a holomorphic vector bundle over the Stein manifold B is Stein, the Oka–Grauert principle for principal bundles [31] (see also [22, Theorem 8.2.1(i), p. 356]) gives a holomorphic isomorphism

$$\Phi : p^*X \xrightarrow{\sim} s^*X.$$

On the zero section, Φ restricts to a holomorphic bundle automorphism $a : X \rightarrow X$ over B . Since Φ is a principal-bundle isomorphism, so is a . Precomposing Φ with the automorphism

$$p^*X \longrightarrow p^*X, \quad (v, x) \longmapsto (v, a^{-1}(x)),$$

we may assume that Φ restricts to the identity on the zero section.

Using the natural identification $p^*X \cong \pi^*V$, write

$$\Phi(v, x) = (v, \tilde{s}(x, v)).$$

Then $\tilde{s}(x, 0) = x$ and

$$\pi(\tilde{s}(x, v)) = s(v).$$

Differentiating $\pi \circ \tilde{s} = s$ in the V -direction along the zero section shows that $d\tilde{s}$ surjects onto $TX/\ker d\pi$, since s is dominating. The remaining vertical directions are supplied by the principal action: if $\mathfrak{g}$ is the Lie algebra of G , then

$$S(x, v, \xi) = \tilde{s}(x, v) \cdot \exp(\xi)$$

is a dominating spray on $(\pi^*V) \oplus (X \times \mathfrak{g})$. $\square$

Remark 8.4. The Stein and principal-bundle hypotheses are used in the lifting step. The corresponding question for an arbitrary elliptic Serre fibration over an elliptic base is open; see [22, Problem 6.4.19].

Proof of Theorem 1.2. Put $\mathcal{R} = \mathcal{R}_3(E, \mathbb{P}^1)$. Bowman associates to E a smooth plane cubic C_E and proves

$$\mathcal{R} \text{ is Oka} \iff \mathbb{P}^2 \setminus C_E \text{ is Oka}$$

[8, Corollary 30]. The right-hand side holds by Theorem 8.1; hence $\mathcal{R}$ is Oka.

Precomposition by translations of E is free, and Bowman proves that the quotient map is a holomorphic principal bundle

$$E \longrightarrow \mathcal{R} \longrightarrow B := \mathcal{R}/E$$

whose base B is Stein [8, Theorems 39 and 40]. Since the fibre E and the total space $\mathcal{R}$ are Oka, [22, Theorem 5.6.5] shows that B is Oka. A Stein Oka manifold is elliptic [22, Proposition 5.6.15]; thus B is elliptic. Applying Theorem 8.3 to the displayed principal E -bundle proves that $\mathcal{R}$ is elliptic. $\square$

Remark 8.5 (Geometric origin of the plane cubic). The cubic C_E is intrinsic to the degree-three mapping space. Building on Namba's lemma [49, Lemma 1.4.1], Bowman obtains a finite unbranched cover of the postcomposition quotient by $(\mathbb{P}^2 \setminus C_E) \times E$; the cover arises from the multiplication-by-three isogeny of E [8, Proposition 28]. This explains geometrically why a plane-cubic complement appears in the criterion above.

8.3. Cubic rational self-maps of the sphere. The proof of Theorem 1.3 first identifies the postcomposition quotient with the base-point-free locus in $\text{Gr}(2, 4)$, and then uses the Bézout matrix and projective inversion to turn that locus into the complement of a cubic hypersurface in $\mathbb{P}^4$. Put

$$V = H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(3)),$$

and let

$$\mathcal{B} = \{W \in \text{Gr}(2, V) : W \text{ is base-point-free on } \mathbb{P}^1\}.$$

For $f = [p : q] \in R_3$, write $W_f = \text{span}(p, q) \in \mathcal{B}$. The key geometric identification is the following.

Theorem 8.6. *There is an algebraic principal $\text{PGL}_2(\mathbb{C})$ -bundle*

$$\text{PGL}_2(\mathbb{C}) \longrightarrow R_3 \longrightarrow \mathbb{P}^4 \setminus D,$$

where

$$D = \{XZW - XV^2 - Y^2W + 2YVZ - Z^3 = 0\} \subset \mathbb{P}^4.$$

Moreover, D is irreducible.

Proof. We proceed in three steps.

Step 1: the postcomposition quotient.

The map

$$\rho : R_3 \longrightarrow \mathcal{B}, \quad [p : q] \longmapsto \text{span}(p, q),$$

is surjective. For $W \in \mathcal{B}$, every ordered basis (p, q) has no common zero and therefore defines a point $[p : q] \in R_3$. Two ordered bases define the same point of R_3 precisely when they differ by a common scalar. Thus $\text{PGL}_2(\mathbb{C})$ acts simply transitively on each fibre of ρ by postcomposition. Equivalently, ρ is the projective frame bundle of the tautological rank-two bundle restricted to $\mathcal{B}$:

$$(8.1) \quad \text{PGL}_2(\mathbb{C}) \longrightarrow R_3 \xrightarrow{\rho} \mathcal{B}.$$

See also [37, §3].

Step 2: the Bézout matrix.

Write

$$\begin{aligned} p &= a_0 z_0^3 + a_1 z_0^2 z_1 + a_2 z_0 z_1^2 + a_3 z_1^3, \\ q &= b_0 z_0^3 + b_1 z_0^2 z_1 + b_2 z_0 z_1^2 + b_3 z_1^3, \end{aligned}$$

and let

$$P_{ij} = a_i b_j - a_j b_i, \quad 0 \leq i < j \leq 3,$$

be the Plücker coordinates of $W = \text{span}(p, q)$. The Plücker embedding identifies $\text{Gr}(2, V)$ with the quadric

$$(8.2) \quad P_{01}P_{23} - P_{02}P_{13} + P_{03}P_{12} = 0$$

in $\mathbb{P}^5$.

The classical Bézout formula associates to W the symmetric matrix

$$(8.3) \quad A(W) = \begin{pmatrix} P_{01} & P_{02} & P_{03} \\ P_{02} & P_{03} + P_{12} & P_{13} \\ P_{03} & P_{13} & P_{23} \end{pmatrix}$$

and gives

$$\det A(W) = \pm \text{Res}(p, q);$$

see [12, Example 4.1 and Proposition 4.2]. Hence

$$W \in \mathcal{B} \iff \det A(W) \neq 0.$$

Set

$$A = \begin{pmatrix} x & y & z \\ y & u & v \\ z & v & w \end{pmatrix}, \quad (x, y, z, u, v, w) = (P_{01}, P_{02}, P_{03}, P_{03} + P_{12}, P_{13}, P_{23}).$$

This linear change of coordinates identifies $\mathbb{P}^5$ with $\mathbb{P}(\text{Sym}_3(\mathbb{C}))$, where $\text{Sym}_3(\mathbb{C})$ denotes the six-dimensional vector space of symmetric 3×3 complex matrices.

Since $P_{12} = u - z$, the Plücker equation becomes

$$(8.4) \quad Q(A) := xw - yv + zu - z^2 = 0.$$

Therefore

$$\mathcal{B} \cong \{[A] \in \mathbb{P}(\text{Sym}_3(\mathbb{C})) : Q(A) = 0, \det A \neq 0\}.$$

Step 3: projective matrix inversion.

Let $\text{adj}(A)$ denote the classical adjugate matrix, characterized by $A \text{adj}(A) = \det(A)I$. On the open subset

$$\{\det A \neq 0\} \subset \mathbb{P}(\text{Sym}_3(\mathbb{C})),$$

the map $\iota : [A] \mapsto [\text{adj}(A)]$ is a biregular involution, since $\text{adj}(\text{adj}(A)) = (\det A)A$. Put $B = \text{adj}(A)$. Since the entries of B read as

$$B_{22} = xw - z^2, \quad B_{13} = yv - zu,$$

we have

$$Q(A) = B_{22} - B_{13}.$$

Thus ι identifies $\mathcal{B}$ with

$$H \setminus \{\det B = 0\}, \quad H := \{B_{22} = B_{13}\} \cong \mathbb{P}^4.$$

Writing

$$B = \begin{pmatrix} X & Y & Z \\ Y & Z & V \\ Z & V & W \end{pmatrix}$$

on H , we obtain

$$\det B = XZW - XV^2 - Y^2W + 2YVZ - Z^3.$$

Hence $\mathcal{B} \cong \mathbb{P}^4 \setminus D$.

It remains to prove that D is irreducible. Set

$$R = \mathbb{C}[Y, Z, V, W], \quad a = ZW - V^2, \quad b = -Y^2W + 2YVZ - Z^3.$$

Then $F = aX + b \in R[X]$. The polynomial a is irreducible in the UFD R (view it as a primitive linear polynomial in W over $\mathbb{C}[Y, Z, V]$), and $a \nmid b$: at $Z = 1$ and $V = W = 0$, one has $a = 0$ but $b = -1$. Hence $\gcd(a, b) = 1$, so F is primitive. It is a nonconstant linear polynomial over $\text{Frac}(R)$, and is therefore irreducible there. Gauss' lemma now gives irreducibility in $R[X]$. $\square$

Proof of Theorem 1.3. By Theorem 8.6, R_3 is a holomorphic principal $\mathrm{PGL}_2(\mathbb{C})$ -bundle over $\mathbb{P}^4 \setminus D$, where D is an irreducible cubic hypersurface. Hence D is not the exceptional union of three hyperplanes in Theorem 1.1, so $\mathbb{P}^4 \setminus D$ is elliptic and, being affine, Stein. Applying Theorem 8.3 to the displayed principal bundle shows that R_3 is holomorphically elliptic. $\square$

Remark 8.7 (Scope of the Bézout reduction). The quotient by base-point-free pencils and its Bézout description exist in every degree. For $d = 2$, the quotient is $\mathbb{P}^2$ minus the conic of pencils with a base point. For $d \geq 2$, the postcomposition quotient of R_d is the base-point-free locus in

$$\mathrm{Gr}(2, H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(d))) \cong \mathrm{Gr}(2, d+1),$$

and the Bézout formula represents the resultant by the determinant of a symmetric $d \times d$ matrix linear in the Plücker coordinates [12, Example 4.1 and Proposition 4.2].

For $d = 3$, two coincidences occur: $\mathrm{Gr}(2, 4) \subset \mathbb{P}^5$ is a quadric hypersurface, and projective inversion of the associated symmetric 3×3 Bézout matrix turns this single quadratic equation into a linear one on the invertible locus. This identifies the quotient with a cubic hypersurface complement in $\mathbb{P}^4$. For $d \geq 4$, the Grassmannian has higher codimension, so there is no single Plücker equation for inversion to linearize. Thus this particular one-equation argument does not directly produce a hypersurface-complement model for the quotient of R_d . This is only a limitation of the present Bézout reduction; it does not rule out a different model in higher degree.

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AI USE DISCLOSURE

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